# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET B

Chapters 3 & 4: **Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Board Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use graph paper wherever a graph is asked.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The pair $x + y = 2$ and $2x + 2y = 4$ has: 1

**(a)** a unique solution **(b)** infinitely many solutions
**(c)** two solutions **(d)** no solution

**2.** The solution of $x + y = 10$ and $x - y = 2$ is: 1

**(a)** $(5, 5)$ **(b)** $(8, 2)$ **(c)** $(4, 6)$ **(d)** $(6, 4)$

**3.** The value of $k$ for which $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$ has no solution is: 1

**(a)** 1 **(b)** 3 **(c)** −2 **(d)** 2

**4.** A student states: "A pair of linear equations can have exactly two solutions." The correct response is: 1

**(a)** Correct
**(b)** Correct for parallel lines
**(c)** Incorrect — it has 0, 1 or infinitely many solutions
**(d)** Cannot be determined

**5.** The line $2x + y = 6$ meets the $y$-axis at: Student A says $(0, 6)$; Student B says $(3, 0)$. Then: 1

**(a)** Student A is correct **(b)** Student B is correct
**(c)** Both are correct **(d)** Both are wrong

**6.** The roots of $x^2 + 5x = 0$ are: 1

**(a)** $-5$ only **(b)** $0, 5$ **(c)** $5, -5$ **(d)** $0, -5$

**7.** If 2 is a root of $x^2 + kx + 12 = 0$, then $k$ is: 1

**(a)** 6 **(b)** −6 **(c)** −8 **(d)** 8

**8.** Which equation has equal roots? 1

**(a)** $x^2 - 16 = 0$ **(b)** $x^2 - 8x + 15 = 0$ **(c)** $x^2 - 8x + 16 = 0$ **(d)** $x^2 + 16 = 0$

**9.** A student states: "If $D = 0$, the quadratic equation has no roots." The correct response is: 1

**(a)** Incorrect — it has two equal real roots **(b)** Correct
**(c)** It has three roots **(d)** Cannot be determined

**10.** "The area of a square of side $(x + 3)$ cm is 49 cm²" is represented by: 1

**(a)** $x^2 - 40 = 0$ **(b)** $x^2 + 6x - 40 = 0$ **(c)** $x^2 + 6x + 9 = 0$ **(d)** $x^2 + 3x - 49 = 0$

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Solve by elimination: $3x - 5y - 4 = 0$ and $9x = 2y + 7$. 2

**12.** Check whether the pair $2x - 3y = 8$ and $4x - 6y = 9$ is consistent. 2

**13.** Solve: $2x^2 - x + \frac{1}{8} = 0$. 2

**14.** For what value of $p$ does $px^2 - 4x + 1 = 0$ have equal roots? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A two-digit number is 4 times the sum of its digits. If 18 is added to it, the digits are reversed. Find the number. 3

**16.** Find the roots of $\frac{1}{x} - \frac{1}{x-2} = 3$, $x \neq 0, 2$. 3

**17.** 10 students of Class X took part in a quiz. If the number of girls is 4 more than the number of boys, find the number of boys and girls. 3

OR

ABCD is a cyclic quadrilateral with $\angle A = 4y + 20$, $\angle B = 3y - 5$, $\angle C = -4x$, $\angle D = -7x + 5$ (degrees). Find the four angles. 3

**18.** The hypotenuse of a right triangle is 25 cm and the difference between its other two sides is 5 cm. Find the two sides. 3

OR

Find the values of $k$ for which $kx^2 - 6x - 2 = 0$ has real roots. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study:** Akhila went to a fair with ₹20 and wanted rides on the Giant Wheel and to play Hoopla. She played Hoopla half as many times as she rode the Giant Wheel. Each ride costs ₹3 and each game ₹4, and she spent ₹20. 5

(i) Form the equations.

(ii) Solve them graphically.

(iii) State the number of rides and games.

OR

A show sold 100 tickets: adult tickets at ₹50 and child tickets at ₹30, collecting ₹4200. 5

(i) Find the number of each type sold.

(ii) If ₹4600 had been collected from 100 tickets, how many adult tickets were sold?

**20.** **Case study:** A man buys a number of pens for ₹80. If he had bought 4 more pens for the same amount, each pen would have cost ₹1 less. 5

(i) Form the equation.

(ii) Find the number of pens.

(iii) Find the cost of each pen.

OR

The product of the digits of a two-digit number is 24. When 18 is subtracted from the number, the digits interchange. Find the number. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**